

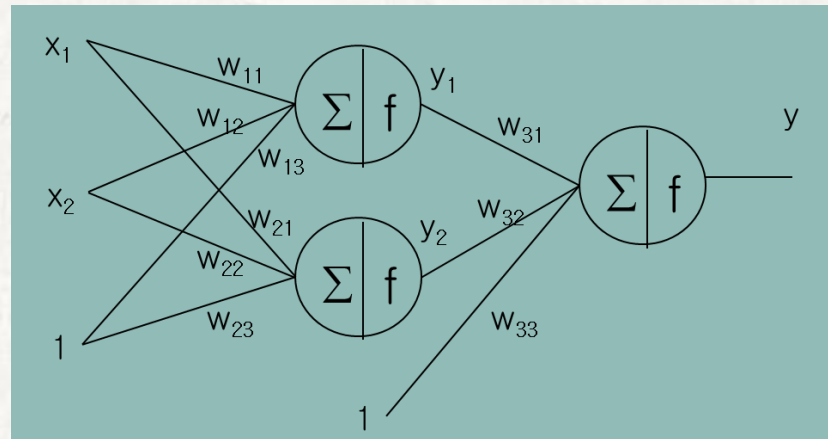
Configuration of Neural Networks

Contents

- Regression
- Binary-Class Classification
- Multi-Class Classification
- Nominal Input

Regression

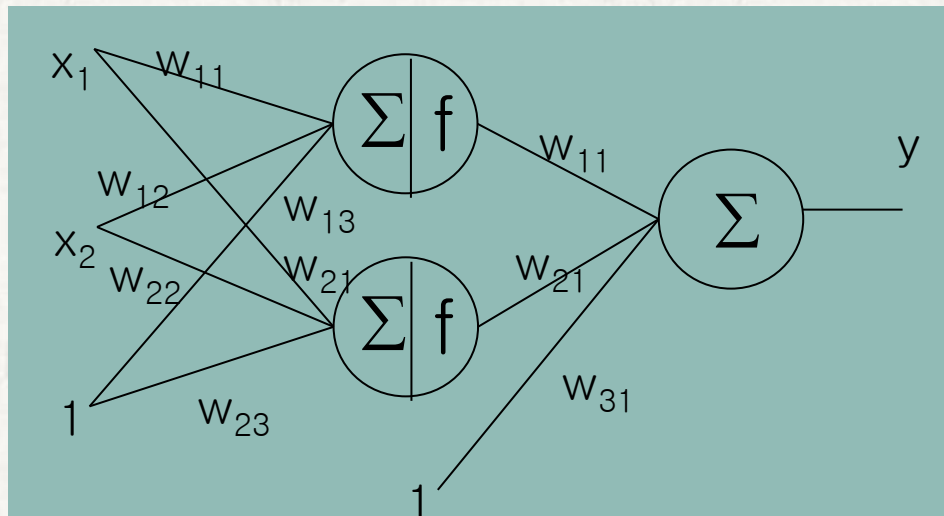
- Following Neural Network is OK for regression?



- Maybe NO!! Why?
- The activation functions produces a value between $[0,1]$

Regression

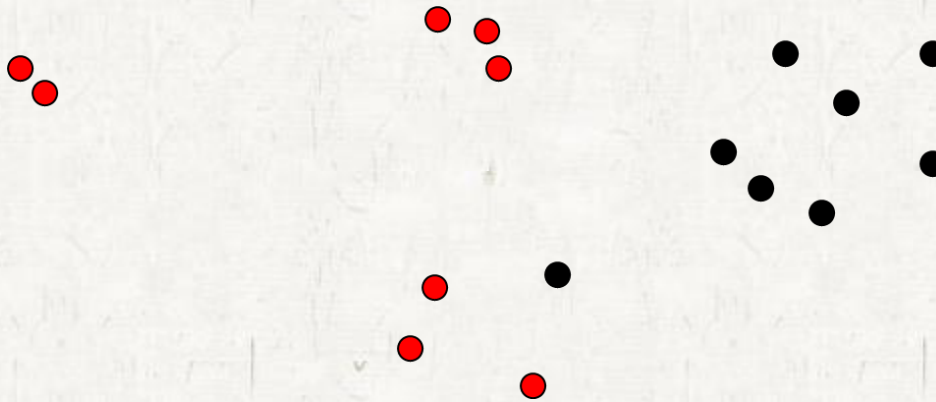
- Solution
 - Use a linear output node



Binary-Class Classification

• You Have Two Problems

$(x_{11}, x_{12}, Red), (x_{21}, x_{22}, Red), (x_{31}, x_{32}, Black), (x_{41}, x_{42}, Red), (x_{51}, x_{52}, Black), \dots$



- P1: NN cannot produce nominal values
- P2: Error Function for training

Binary-Class Classification

● P1: Handling Nominal Values

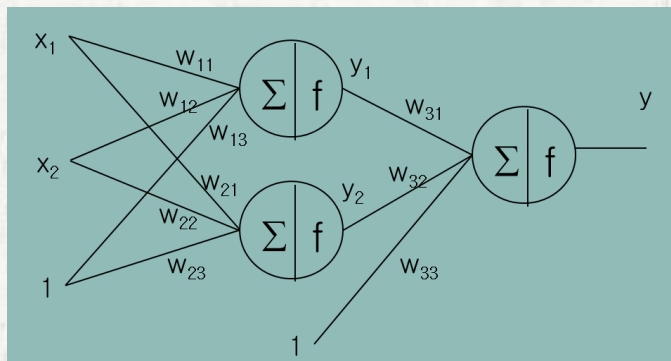
- Use 0 and 1 for class labels

$(x_{11}, x_{12}, Red), (x_{21}, x_{22}, Red), (x_{31}, x_{32}, Black), (x_{41}, x_{42}, Red), (x_{51}, x_{52}, Black), \dots$



$(x_{11}, x_{12}, 1), (x_{21}, x_{22}, 1), (x_{31}, x_{32}, 0), (x_{41}, x_{42}, 1), (x_{51}, x_{52}, 0), \dots$

- Use Sigmoid



Binary-Class Classification

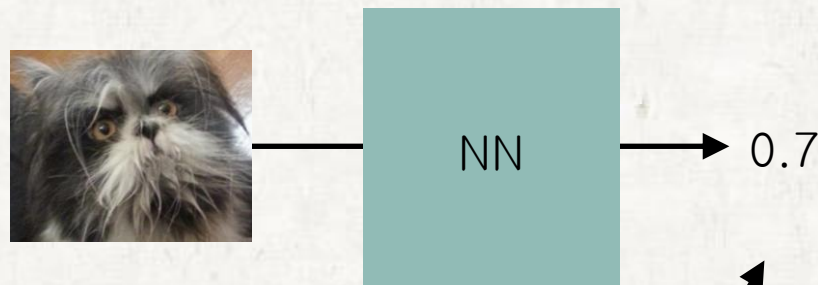
- But.. There is a problem

Training Data



...

Test Data



I call this “**Label**”, then How can we interpret this?

Binary-Class Classification

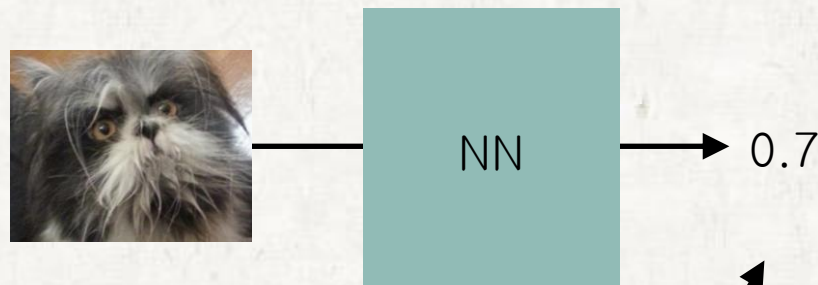
- But.. There is a problem

Training Data



...

Test Data



Let's **regard** this as “**Probability** of Dog”, then it is easy to interpret

Binary-Class Classification

- But.. There is another problem !
 - Output of Classification → Probability
 - We need to describe the training process using “probability”

Find w so that NN correctly predicts all training data



Find w which maximizes the probability that NN correctly predicts all training data



Find w which maximizes the following:

$$\left(\prod_{(x,1) \in \text{Data}} NN(x; w) \right) \times \left(\prod_{(x,0) \in \text{Data}} (1 - NN(x; w)) \right)$$

Binary-Class Classification

● Cross Entropy

$$\begin{aligned}
 & \operatorname{argmax}_w \left(\prod_{(x,1) \in \text{Data}} NN(x; w) \times \prod_{(x,0) \in \text{Data}} (1 - NN(x; w)) \right) \\
 &= \operatorname{argmax}_w \log \left(\prod_{(x,1) \in \text{Data}} NN(x; w) \times \prod_{(x,0) \in \text{Data}} (1 - NN(x; w)) \right) \\
 &= \operatorname{argmax}_w \left(\sum_{(x,1) \in \text{Data}} \log NN(x; w) + \sum_{(x,0) \in \text{Data}} \log(1 - NN(x; w)) \right) \\
 &= \operatorname{argmax}_w \left(\sum_{(x,1) \in \text{Data}} y \log NN(x; w) + (1 - y) \log(1 - NN(x; w)) + \sum_{(x,0) \in \text{Data}} y \log NN(x; w) + (1 - y) \log(1 - NN(x; w)) \right) \\
 &= \operatorname{argmax}_w \left(\sum_{(x,y) \in \text{Data}} y \log NN(x; w) + (1 - y) \log(1 - NN(x; w)) \right) \\
 &= \operatorname{argmin}_w \left(- \sum_{(x,y) \in \text{Data}} y \log NN(x; w) + (1 - y) \log(1 - NN(x; w)) \right)
 \end{aligned}$$

Binary-Class Classification

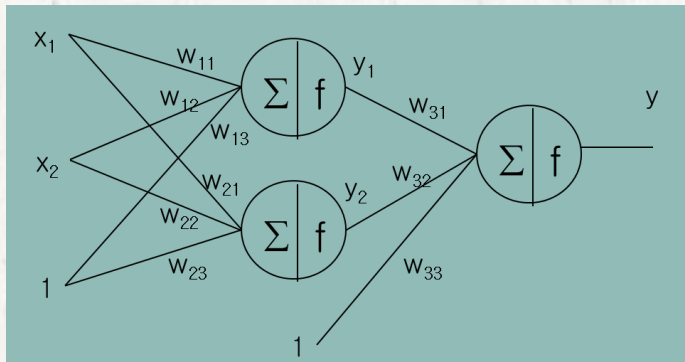
Summary

① Preprocessing

$(x_{11}, x_{12}, 1), (x_{21}, x_{22}, 1), (x_{31}, x_{32}, 0), (x_{41}, x_{42}, 1), (x_{51}, x_{52}, 0), \dots$

② Sigmoid at output node

③ Cross Entropy



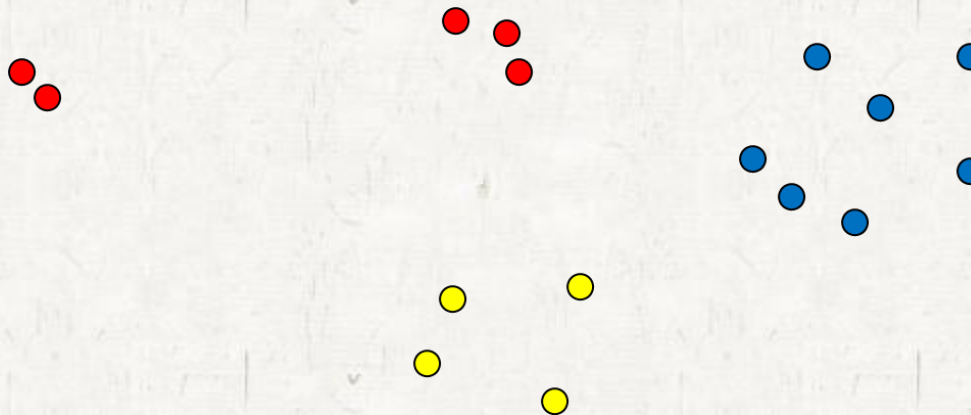
$$E = - \sum_{n=1}^N (t_n \log(y_n) + (1 - t_n) \log(1 - y_n))$$

where $t_n \in \{0,1\}$ and $y_n \in [0,1]$

Multi-Class Classification

● Problem

$(\mathbf{x}_1, \text{Red}), (\mathbf{x}_2, \text{Yellow}), (\mathbf{x}_3, \text{Blue}), (\mathbf{x}_4, \text{Red}), (\mathbf{x}_5, \text{Blue}), \dots$



● Easy...

Multi-Class Classification

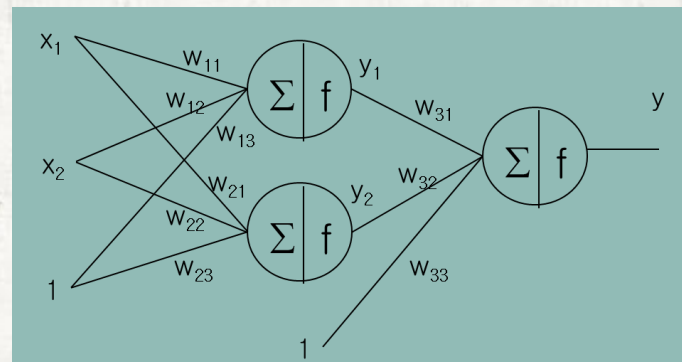
- Nominal Value Handling: Linear conversion of class labels

$(x_{11}, x_{12}, Red), (x_{21}, x_{22}, Yellow), (x_{31}, x_{32}, Blue), (x_{41}, x_{42}, Red), (x_{51}, x_{52}, Blue), \dots$



$(x_{11}, x_{12}, 1), (x_{21}, x_{22}, 0.5), (x_{31}, x_{32}, 0), (x_{41}, x_{42}, 1), (x_{51}, x_{52}, 0), \dots$

- Use Sigmoid at output node
- Prediction



$$class(\mathbf{x}) = \begin{cases} 1 & NN(\mathbf{x}) \geq 2/3 \\ 0.5 & 2/3 \geq NN(\mathbf{x}) \geq 1/3 \\ 0 & Otherwise \end{cases}$$

Multi-Class Classification

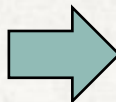
- Not Good... why?
 - There is no order between Red, Yellow, Blue
 - They are just names. We cannot say that $\text{Red} > \text{Yellow} > \text{Blue}$
 - Linear conversion changes the original problem.

Red	→	1.0
Yellow	→	0.5
Blue	→	0.0

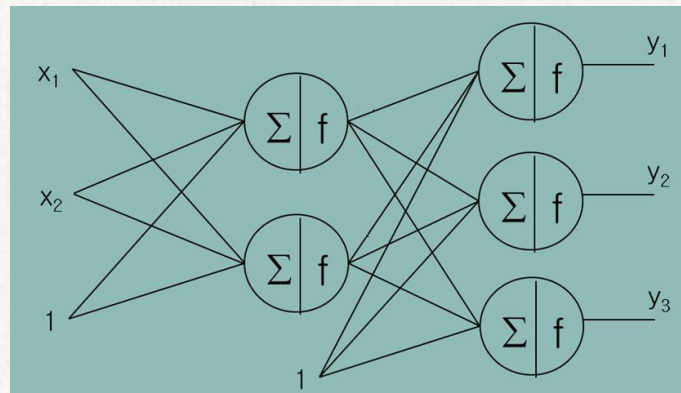
Multi-Class Classification

Then?

① Create virtual outputs

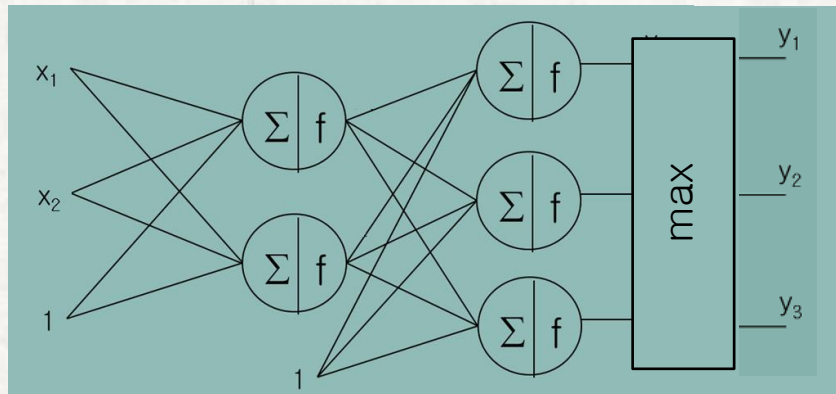
$(x_{11}, x_{12}, Red),$		$(x_{11}, x_{12}, 1, 0, 0),$
$(x_{21}, x_{22}, Yellow),$		$(x_{21}, x_{22}, 0, 1, 0),$
$(x_{31}, x_{32}, Blue),$		$(x_{31}, x_{32}, 0, 0, 1),$
$(x_{41}, x_{42}, Red),$		$(x_{41}, x_{42}, 1, 0, 0),$
$(x_{51}, x_{52}, Blue),$		$(x_{51}, x_{52}, 0, 0, 1),$
...		...

② Place nodes at the output layer as many as outputs



Multi-Class Classification

③ Choose the maximum



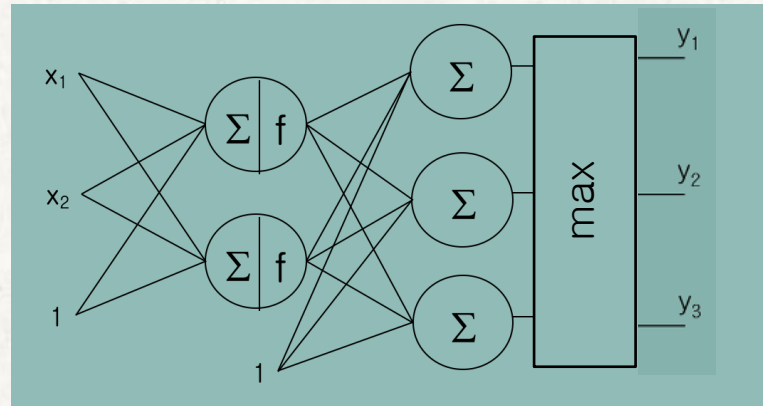
Redefine “max” to return the position of the maximum value

$$(1, 0, 0) = \max(10, 5, 2)$$

Multi-Class Classification

③ Choose the maximum (con'd)

Since the sigmoid is monotonically increasing, we have the same output if we remove the sigmoid activations.



But.. “max” is not differentiable.

Let's find other function similar to “max” but differentiable.

Multi-Class Classification

③ Choose the softmax instead of max

$$(y_1, y_2, y_3) = \text{softmax}(x_1, x_2, x_3)$$

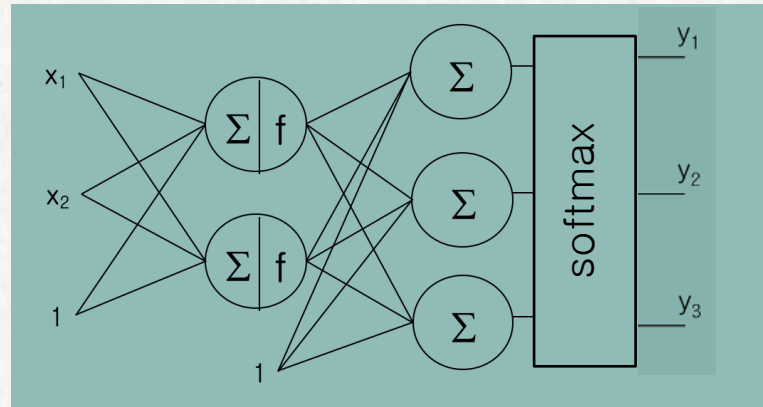
$$y_k = \frac{e^{x_k}}{\sum_{i=1}^n e^{x_i}}$$

y_1	y_2	y_3
0.301	0.332	0.367
0.090	0.245	0.665
0.042	0.114	0.844
0.017	0.047	0.936
0.000	0.000	1.000
0.000	0.000	1.000

x_1	x_2	x_3
1	1.1	1.2
1	2	3
1	2	4
1	2	5
1	2	10
1	2	20

Multi-Class Classification

Loss Function



$$E = \sum_{n=1}^{Data} \sum_{k=1}^{Class} -t_{nk} \log(y_{nk})$$

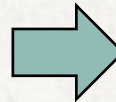
$$\text{Hmm?? } -(t_n \log(y_n) + (1 - t_n) \log(1 - y_n))$$

Multi-Class Classification

Summary

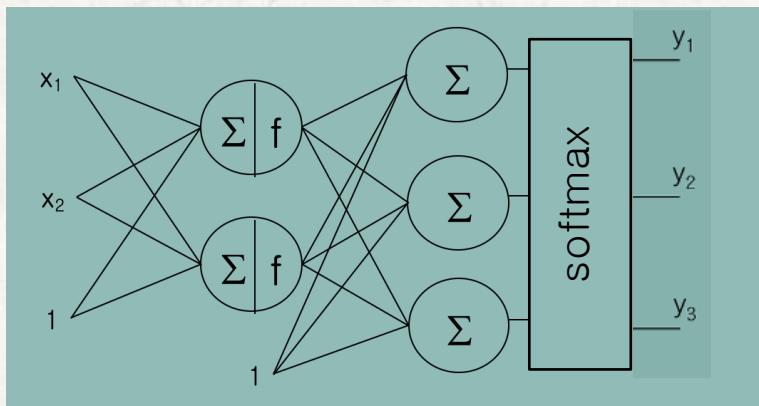
① Create virtual outputs

$(x_{11}, x_{12}, Red),$
 $(x_{21}, x_{22}, Yellow),$
 $(x_{31}, x_{32}, Blue),$
 $(x_{41}, x_{42}, Red),$
 $(x_{51}, x_{52}, Blue),$
 ...



$(x_{11}, x_{12}, 1, 0, 0),$
 $(x_{21}, x_{22}, 0, 1, 0),$
 $(x_{31}, x_{32}, 0, 0, 1),$
 $(x_{41}, x_{42}, 1, 0, 0),$
 $(x_{51}, x_{52}, 0, 0, 1),$
 ...

② Use softmax



③ Use cross entropy

$$E = \sum_{n=1}^{Data} \sum_{k=1}^{Class} -t_{nk} \log(y_{nk})$$

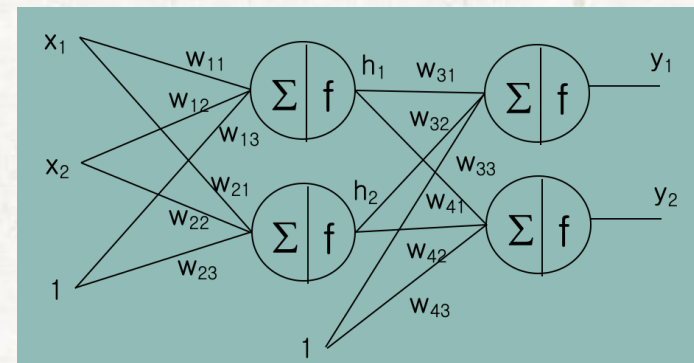
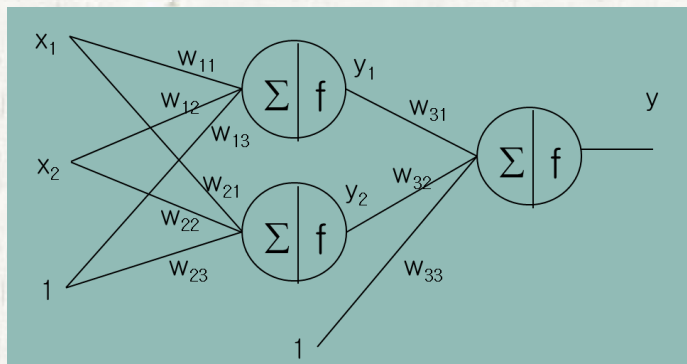
Multi-Class Classification

● Cross Entropy for Multi-Class

(x_{11}, x_{12}, Red)
 (x_{21}, x_{22}, Red)
 $(x_{31}, x_{32}, Black)$
 (x_{41}, x_{42}, Red)
 $(x_{51}, x_{52}, Black)$

$(x_{11}, x_{12}, 1)$
 $(x_{21}, x_{22}, 1)$
 $(x_{31}, x_{32}, 0)$
 $(x_{41}, x_{42}, 1)$
 $(x_{51}, x_{52}, 0)$

$(x_{11}, x_{12}, 1, 0)$
 $(x_{21}, x_{22}, 1, 0)$
 $(x_{31}, x_{32}, 0, 1)$
 $(x_{41}, x_{42}, 1, 0)$
 $(x_{51}, x_{52}, 0, 1)$



$$-(t_n \log(y_n) + (1 - t_n) \log(1 - y_n)) \quad -(t_{n1} \log(y_{n1}) + t_{n2} \log(y_{n2})) = - \sum_{k=1}^{Class} t_{nk} \log(y_{nk})$$

Nominal Inputs

- What if you have categorical inputs

- Two inputs and one output

$$x_1 \in R$$

$$x_2 \in \{Red, Yellow, Blue\}$$

$$y \in \{0,1\}$$

- Create a new input variable for each categorical value

$$x_2 = \begin{cases} 1 & \text{if original } x_2 \text{ is Yellow} \\ 0 & \text{Otherwise} \end{cases}$$

$$x_3 = \begin{cases} 1 & \text{if original } x_2 \text{ is Red} \\ 0 & \text{Otherwise} \end{cases}$$

$$x_4 = \begin{cases} 1 & \text{if original } x_2 \text{ is Blue} \\ 0 & \text{Otherwise} \end{cases}$$

(0.1, Red, 0)

(0.2, Blue, 1)

(0.3, Yellow, 0)

(0.4, Red, 1)



(0.1, 1, 0, 0, 0)

(0.2, 0, 0, 1, 1)

(0.3, 0, 1, 0, 0)

(0.4, 1, 0, 0, 1)

Summary

Problem	Activation Function		Loss function
	Hidden Layer	Output Layer	
Regression	ReLU	Linear	MSE
2-class Classification	ReLU	Sigmoid	CE
Multi-class Classification	ReLU	Softmax	CE